
VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590018



"VTU Internship Report"

ON

"Android App Development using Generative AI"

Submitted in partial fulfilment of the requirements for the award of the degree in the 8th semester.

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

BY

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1ME22CS092

Under the Guidance of

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MS ENGINEERING COLLEGE

*Approved By AICTE New Delhi, Affiliated to VTU, Belagavi, Karnataka
NAAC Accredited and An ISO 9001:2015 Certified Institution*

**Department of Computer Science and Engineering
Academic Year 2025 – 2026**



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the Internship Report entitled "**Android App Development Using GenAI – GenAI- Namma-Pustaka (Education)**" is a Bonafide work carried out by **Nalin Kumar PS(1ME22CS092)** in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi**, during the year **2025–2026**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report, deposited in the department library. This internship report has been approved as it satisfies the academic requirements in respect of internship work prescribed for the Bachelor of Engineering Degree.

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2. External Examiner Name: _____

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



DECLARATION

I, Nalin Kumar Ps(1ME22CS092), Students of 8th Semester B.E Computer science and Engineering, **MS Engineering College**, Bangalore, Declare that this Internship work Entitled “**Android App Development Using Gen AI**” is the original work carried by me under the guidance and supervision provided during the internship period in the partial fulfilment of the course requirements for the award of the Degree in **Bachelor of Engineering in Computer Science and Engineering** by **Visvesvaraya Technological university, Belagavi** during the year 2025-2026

SIGNATURE OF THE STUDENT

ACKNOWLEDGEMENT

I would like to express my deep sense of gratitude to our Principal **Dr. N Ranapratap Reddy, M S Engineering College**, Bengaluru for his inspiration and academic support by providing good facilities to complete the internship.

I extend special in-depth, heartfelt, and sincere gratitude to **Dr. Malatesh S H**, Head of CSE Department, for his wholehearted support in completion of this internship.

I would like to express my heartfelt gratitude to **Mr. Tirumal Mutalikdesai**, External Internship Guide at **MINDMATRIX**, for giving me the opportunity, guidance, and support to complete my internship.

My sincere thanks to my internship coordinator **Ms. Harshitha S P**, Assistant Professor, Dept. of CSE, for guiding and giving me timely advice and suggestions in the successful completion of the internship.

I am highly indebted to my Internal Internship Guide **Dr.Dipti Patnayak** for guiding and giving me timely advice and suggestions in the successful completion of the internship.

I thank all the staff members for supporting me in completion of the internship. Also, I am thankful to my parents and all my dear friends who have directly or indirectly helped me.

Finally, I express gratitude to all those who knowingly or unknowingly helped me in successful completion of this internship.

NALIN KUMAR PS
(1ME22CS092)

Internship Offer Letter

To,
Nalin Kumar ps

Date : 8 December 2025

Subject: Internship Offer Letter – Android App Development using Gen AI

Dear Nalin Kumar ps,

Thank you for exploring internship opportunities with **MindMatrix.io**, a product of **CL Infotech Pvt. Ltd.**, through the **VTU Internship Portal**. We are pleased to offer you an internship with our organization.

This internship will run for **13 to 15 weeks** as per VTU Guidelines. The Internship will provide you with hands-on exposure through guided tasks, structured modules, and practical projects on the **MindMatrix Platform** (<https://lms.mindmatrix.io>).

During the internship, you will be engaged in product application development, supporting feature implementation, testing, preparing documentation, and cross-functional collaboration. These activities will require you to apply your technical knowledge and demonstrate your professional skills effectively.

All **internship policies and professional guidelines** will be provided in the **Internship Manual**, which will be shared with you on the day of program commencement.

As part of the process, you are required to:

- Print this offer letter and fill in the **Faculty Coordinator details** below.
- Obtain **HoD/Internship Coordinator signature with seal**.
- Email the scanned copy back to us within **one week** of receiving it.

For onboarding instructions, please refer to the attached **"Onboarding Steps"** document. We look forward to your participation and wish you a productive internship.

Warm regards,

Tirumal Mutalikdesai

Program Coordinator, MindMatrix.io

contact@mindmatrix.io | 96115 46444

☐ Authorization from Internship In-charge Faculty Or HoD

Name : Dr. Mallesh S.H Designation: professor & HOD

Contact No.: 9632184822 Email: profmallesh@msc.ac.in Dept: computer Science & Engineering

Sign and Seal of HoD: _____

Professor & Head

Department of Computer Science & Engineering
MS Engineering College

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CHAPTER 1

ABOUT THE ORGANIZATION

MindMatrix is a specialized ed-tech and software ecosystem organization founded by Mr. Sujit Kumar. The organization focuses on bridging the gap between academic learning and the professional software industry through outcome-driven skill development, AI-focused training, and structured problem-solving approaches. MindMatrix provides hands-on experience in Android application development, Generative AI integration, Firebase services, and modern software engineering practices. Through project-based internships, mentorship programs, and real-world application development, the organization helps engineering students gain practical industry exposure and professional skills. The company also promotes the “Campus to Corporate” framework, preparing students for successful careers in the software and technology industry.

1.1 History

MindMatrix traces its origins back to 2018, when the organization began its journey as a small initiative founded by a team passionate about technology and innovation. Driven by the vision of bridging the gap between academic learning and industry requirements, the founders established a platform focused on practical software development and technical training. The organization was formally established in the year 2021, marking an important milestone in its growth and development journey. Since its inception, MindMatrix has continuously expanded its technical training programmes, Android application development initiatives, Generative AI learning activities, and software engineering services, gradually establishing itself as a recognized and credible organization in the field of technology education and software development.

1.2 Company Strategy

MindMatrix operates with a clear strategic framework centered around innovation, technical excellence, and student transformation. The organization emphasizes practical learning methodologies that prepare students for professional careers in software engineering and artificial intelligence.

Purpose

The primary purpose of MindMatrix is to create a strong ecosystem where students can gain practical industry experience, develop technical expertise, and become industry-ready professionals capable of solving real-world problems using modern technologies.

Mission

The mission of MindMatrix is to provide high-quality technical mentorship, practical software development training, and outcome-driven internship programmes that help students transition from academic learning to professional software engineering environments.

Vision

The organization envisions becoming a leading platform for technology education and AI-driven innovation by empowering students with practical skills, modern development practices, and professional industry exposure.

Core Values

The core values followed by MindMatrix include:

- Innovation and continuous learning
- Professional ethics and integrity
- Practical implementation-oriented education
- Collaboration and teamwork
- Creativity and problem-solving
- Student-focused mentorship and growth

Goals

To transform its vision into practical and measurable achievements, MindMatrix has established the following organizational goals:

- To continuously enhance technical expertise and practical learning methodologies in the domains of Android application development, Generative AI, cloud technologies, and software engineering while building a strong student and professional community.
 - To understand the evolving requirements of students, developers, and industry trends and provide industry-oriented training programmes and innovative software solutions that effectively address modern technological challenges.
-

Research and Innovation Department

This department focuses on Generative AI research, machine learning applications, AI orchestration frameworks, and modern technology integration for building intelligent software systems.

1.6 Company Contact Information

Detail	Information
Registered Name	MindMatrix.io (CL Infotech Pvt. Ltd.)
Address	MindMatrix, #651, Kiran Arcade, 4th Floor, 13th Cross, 27th Main Road, 1st Sector, HSR Layout, Bangalore – 560102
Mobile	+91 96115 46444
Email	program_coordinator@mindmatrix.io
Website	Mindmatrix LMS – Personalized Learning & Career Mentorship

CHAPTER 2

OBJECTIVES OF INTERNSHIP

The internship undertaken at MindMatrix focused on Android App Development using Generative AI and modern software development technologies. The primary aim of the internship was to bridge the gap between academic learning and real-world industry practices by providing practical exposure to Android application development, AI integration, cloud technologies, and realtime software engineering workflows. The internship emphasized hands-on project development, problem-solving, professional communication, and industry-oriented technical learning under the guidance of experienced mentors.

2.1 Primary Objectives

The internship programme was designed around several important objectives that guided the technical and professional learning process throughout the internship period. These objectives include:

- **To Apply Academic Knowledge in Practical Development:** The internship aimed to help apply theoretical concepts learned during academics — including programming, databases, software engineering, and mobile application development — into practical Android applications and realtime software projects.
 - **To Develop Proficiency in Android Development Technologies:** A major objective was to gain practical experience with Android Studio, Kotlin, Jetpack Compose, Firebase, APIs, and modern UI frameworks used in professional Android application development.
 - **To Understand Generative AI Integration:** The internship focused on understanding how Generative AI technologies and AI APIs can be integrated into Android applications to create intelligent and interactive software solutions.
 - **To Work on Realtime Industry Projects:** Another key objective was to contribute
-

to practical software projects that simulate real industry workflows, helping develop confidence in handling project implementation, debugging, testing, and deployment activities.

- **To Improve Problem-Solving and Analytical Thinking:** The internship encouraged logical thinking, debugging skills, and structured problem-solving approaches while developing software applications and handling technical challenges.

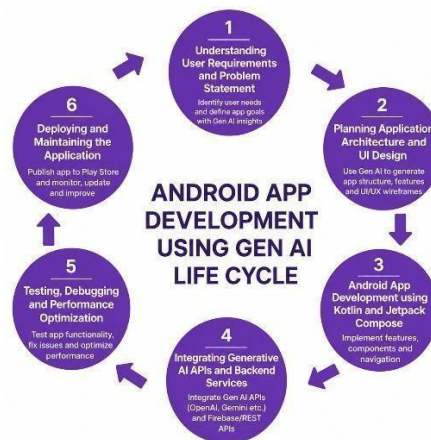


Figure 1 : Android App Development Using Gen AI Life Cycle

2.2 Technical Learning Objectives

Beyond the primary objectives, the internship was designed to achieve several technical learning goals across Android development, AI integration, backend connectivity, and cloud deployment.

Android Application Development

- To learn Android application development using Kotlin and Jetpack Compose.
 - To understand Android activity lifecycle management, navigation systems, UI components, and responsive application design.
 - To develop practical knowledge of Firebase Authentication, Firestore Database, and realtime data synchronization.
-

CHAPTER 3

LEARNING EXPERIENCES

The internship at MindMatrix was an enriching and industry-oriented learning experience that provided practical exposure beyond traditional classroom education. Throughout the internship period, continuous involvement in Android application development, Generative AI assisted workflows, Firebase integration, realtime database management, frontend engineering, and cloud deployment activities helped in building a strong technical and professional foundation. The internship introduced various technical challenges that enhanced logical thinking, debugging abilities, creativity, adaptability, and structured problem-solving skills. The learning experience covered multiple modern software development domains including Android development using Kotlin and Jetpack Compose, Firebase backend integration, responsive UI design, realtime synchronization systems, and AI-assisted software engineering practices. Under the guidance of experienced mentors at MindMatrix, valuable insights into professional software development workflows were gained. The successful completion of multiple Android applications and realtime projects significantly improved technical confidence, communication abilities, teamwork skills, and overall industry readiness required for modern software engineering environments.

3.1 Orientation and Initial Exposure

The internship commenced with a structured onboarding and orientation phase during which the work culture, technical expectations, learning objectives, project workflows, and organizational practices followed at MindMatrix were introduced. This initial phase played an important role in establishing a clear understanding of the internship structure and technical learning roadmap.

- **Understanding the Work Environment:**

The initial days of the internship focused on understanding the organization's learning ecosystem, project workflows, coding practices, communication methods, and development standards. This provided a clear understanding of how realtime

1. Functional Testing

Functional testing was performed to verify all application features.

Tested Functions:

- Adding books
- Updating book details
- Deleting books
- Borrowing books
- Returning books
- QR code scanning
- Review and rating submission
- Student information management

All functionalities were tested successfully.

2. User Interface Testing

UI testing was conducted to check:

- Screen responsiveness
- Button functionality
- Navigation flow
- Layout alignment
- User interaction

The application provided a smooth and user-friendly interface.

3. Database Testing

Room Database operations were tested to ensure:

- Proper data insertion
- Accurate data retrieval
- Successful updates
- Correct deletion of records

Database performance was verified successfully.

4. QR Scanner Testing

The QR code scanning feature was tested using different QR codes to ensure:

- Accurate scanning
- Fast detection
- Proper integration with the application

The scanner worked correctly during testing.

5. Performance Testing

Performance testing was performed to evaluate:

- Application speed
- Smooth navigation
- Data loading performance
- Memory usage

The application showed stable performance during execution.

Challenges Faced During Deployment

Some technical issues were encountered during development and deployment, including:

- Gradle dependency conflicts
- KSP and Kotlin version compatibility issues
- Database configuration errors
- Build and synchronization issues

These issues were resolved by updating dependency versions and correcting Gradle configurations.

Personal and Professional Learning Outcomes

1. Problem Solving Skills

Improved ability to analyze technical problems and identify effective solutions during development.

2. Project Development Experience

Understood the complete software development lifecycle including:

- Requirement analysis
- Design
- Development
- Testing
- Deployment

3. Time Management

Learned how to manage project tasks and complete development activities within the project timeline.

4. UI/UX Understanding

Developed better understanding of creating user-friendly and visually attractive mobile applications.

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Snapshots of Project 1

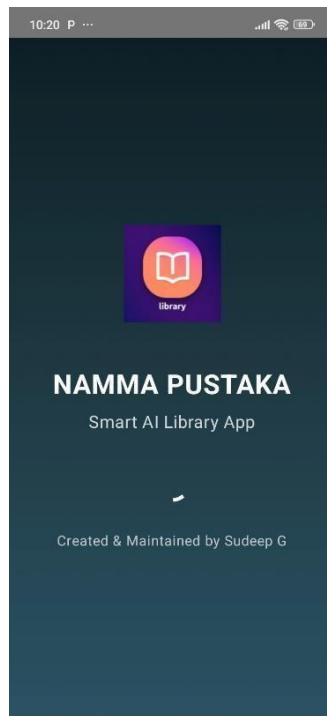


Figure 1: Login Page

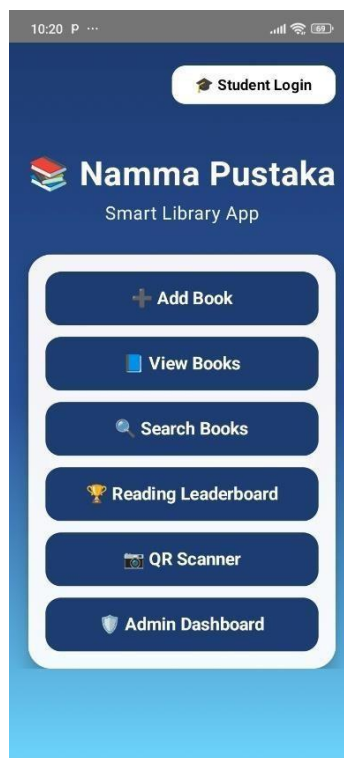
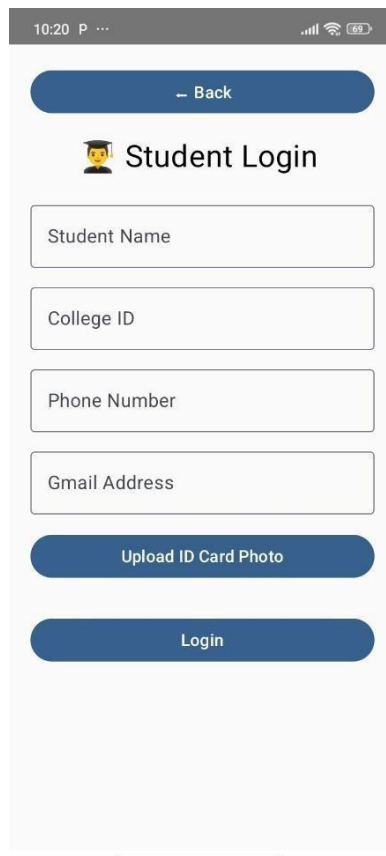
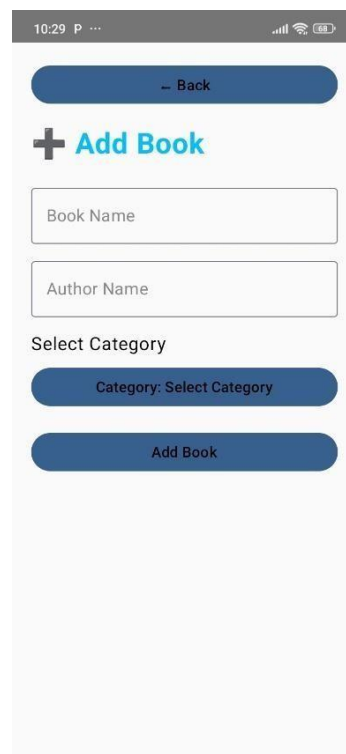


Figure 2: Dashboard Page



The screenshot shows the 'Student Login' screen of an Android application. At the top, there is a status bar with the time '10:20 P' and battery level '68'. Below the status bar is a blue button with a white arrow and the text 'Back'. The title 'Student Login' is displayed in black text next to a student icon. Below the title are four white input fields with rounded corners, labeled 'Student Name', 'College ID', 'Phone Number', and 'Gmail Address'. Below these fields are two blue buttons with white text: 'Upload ID Card Photo' and 'Login'.

Figure 3: Student Login



The screenshot shows the 'Add Book' screen of an Android application. At the top, there is a status bar with the time '10:29 P' and battery level '68'. Below the status bar is a blue button with a white arrow and the text 'Back'. The title '+ Add Book' is displayed in blue text next to a plus icon. Below the title are two white input fields with rounded corners, labeled 'Book Name' and 'Author Name'. Below these fields is the text 'Select Category' followed by a blue button with white text 'Category: Select Category'. At the bottom is a blue button with white text 'Add Book'.

Figure 4: ADD Book

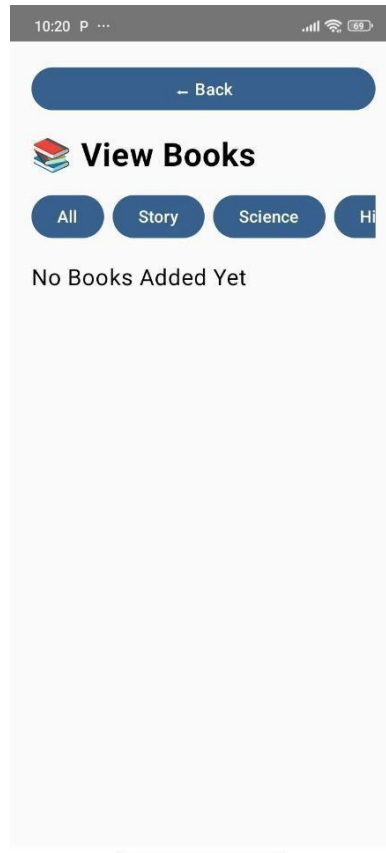


Figure 5: View Book

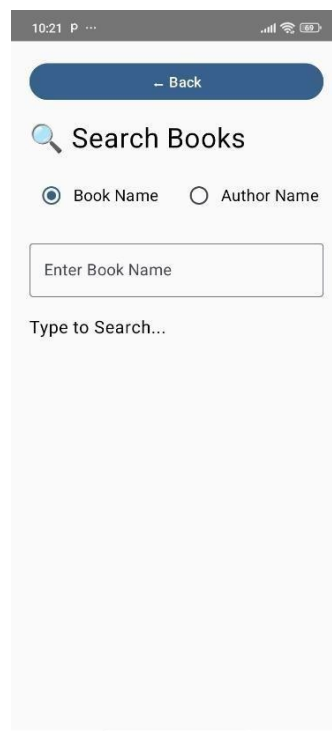


Figure 6: Search Book

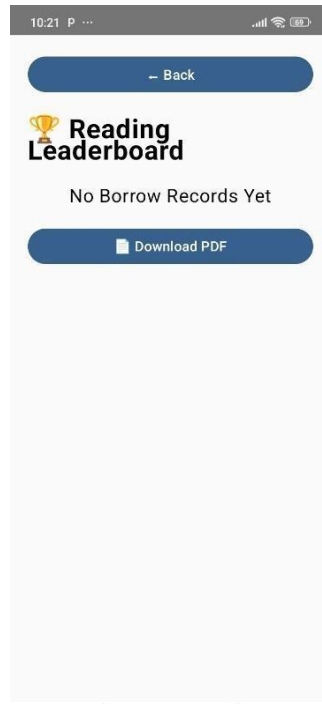


Figure 7: Reading Leaderboard

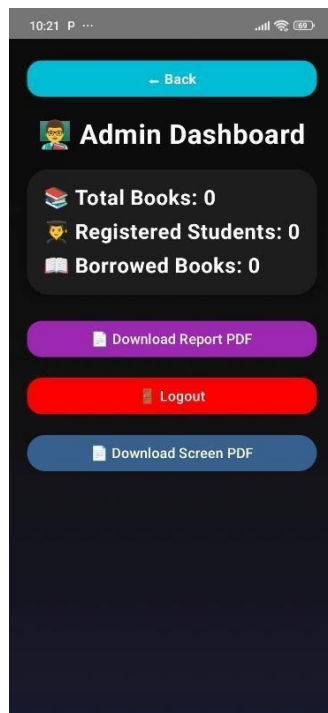


Figure 8: Admin Dashboard

respect, and ethical work practices within organizational and software engineering environments.

- The internship also strengthened long-term career interest and motivation towards Android application development, Generative AI technologies, cloud computing, and modern software engineering practices.

Overall, the internship experience at MindMatrix played a major role in improving not only technical competence but also professional maturity, adaptability, communication skills, teamwork understanding, and career readiness required for future academic projects and software engineering opportunities.



Figure 9: Professional Growth & Career Readiness

CHAPTER 4

LEARNING OUTCOMES

The internship at MindMatrix resulted in a comprehensive set of learning outcomes that significantly improved technical knowledge, professional competence, problem-solving abilities, and personal confidence. The experience of working on realtime Android application development projects, Firebase backend systems, Generative AI assisted workflows, and modern software engineering practices provided practical exposure far beyond traditional academic learning. Unlike classroom-based theoretical concepts, the internship outcomes were achieved through direct implementation, debugging activities, project development, collaborative learning, and continuous interaction with professional development environments. This chapter summarizes the major learning outcomes achieved during the internship and reflects the overall growth in Android development, Generative AI integration, cloud technologies, and professional software engineering practices.

4.1 Technical Outcomes

The most measurable outcomes of the internship were achieved in the domain of technical learning and practical software engineering skills. Throughout the internship, substantial improvement was achieved across Android development, Firebase integration, frontend engineering, cloud technologies, and AI-assisted development methodologies.

4.1.1 Proficiency in Python and the Data Science Ecosystem

By the conclusion of the internship, a strong and confident command of Python had been developed — not merely as a scripting language but as a professional tool for building structured, modular, and production-oriented software. Libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Plotly became part of a reliable and frequently used toolkit, applied across multiple project contexts with increasing speed and efficiency. The ability to write clean, well-documented Python code that adheres to professional coding standards was a particularly valued outcome of the experience.

4.1.1 Proficiency in Android Development using Kotlin and Jetpack Compose

By the completion of the internship, strong practical understanding of Android application development using Kotlin and Jetpack Compose was achieved. Multiple Android applications were developed during the internship including Greeting Card App, Dice Roller App, Lemonade App, Business Card App, Art Space App, and Siri-Dhanya Hub App. These projects significantly improved understanding of composable architecture, responsive UI design, navigation systems, state management, debugging techniques, and modular Android application development practices.

4.1.2 Firebase Backend Integration and Realtime Database Management

One of the major technical outcomes of the internship was gaining practical experience in Firebase Authentication and Firebase Firestore integration. Hands-on implementation of realtime cloud database synchronization, secure authentication systems, user session management, and backend connectivity workflows established a strong foundation in cloud-supported Android application development.

4.1.3 Generative AI Assisted Development

The internship also provided valuable practical exposure to Generative AI assisted development methodologies. Working with AI-supported implementation workflows improved understanding of prompt engineering, AI-assisted code generation, intelligent debugging support, frontend generation assistance, and productivity enhancement techniques used within modern software engineering environments.

4.1.4 Frontend Engineering and Responsive UI Development

Another important technical outcome was the development of responsive Android frontend engineering skills using Jetpack Compose and Material Design principles. Practical implementation of interactive user interfaces, navigation workflows, modular composable screens, animations, and responsive layouts significantly improved frontend architecture understanding and Android UI development capabilities.

4.1.5 GitHub Version Control and Collaborative Development

- The internship provided practical understanding of how modern organizations utilize Android applications, cloud technologies, Firebase systems, and AI-assisted workflows to build intelligent and scalable software solutions.
 - Exposure to Android frontend engineering, realtime Firebase synchronization, and responsive UI architecture expanded technical interest towards modern mobile application development and cloud-supported software systems.
 - Successful completion of multiple Android applications and the Siri-Dhanya Hub App project significantly improved confidence in handling complex Android implementation tasks independently.
 - The internship highlighted the importance of building practical project portfolios and realtime implementation experience in addition to academic performance within competitive software engineering domains.
 - Working with Kotlin, Jetpack Compose, Firebase Authentication, Firebase Firestore, GitHub, and Generative AI tools improved awareness regarding current industry standards and modern Android engineering practices.
 - The learning experience strengthened motivation towards exploring advanced technologies related to Android engineering, cloud computing, AI-assisted development systems, and intelligent mobile applications.
 - Exposure to realtime project workflows, debugging activities, Firebase integration systems, and collaborative software engineering practices improved understanding of professional industry expectations.
 - The internship encouraged continuous self-learning through Android documentation, Firebase developer resources, GitHub repositories, technical blogs, and AI-assisted development tools.
 - The experience also emphasized the importance of adaptability within rapidly evolving software engineering environments where frameworks, tools, and development methodologies continuously change.
-

- Most importantly, the internship strengthened long-term career interest and confidence towards pursuing opportunities related to Android application development, Generative AI integration, cloud technologies, and modern software engineering domains.

From an academic perspective, the internship experience will positively contribute towards future projects, technical coursework, and advanced implementation-oriented learning activities. From a professional perspective, the practical exposure, technical confidence, debugging experience, and software development knowledge gained during the internship have established a strong foundation for future software engineering and Android development opportunities.

4.7 Overall Reflection on the Internship Experience

The internship at MindMatrix was not only a technical learning experience but also an important phase of professional growth, practical transformation, and career preparation. Reflecting on the internship as a whole, the experience provided much greater value than simply completing assigned implementation tasks or fulfilling academic internship requirements. It offered practical exposure to realtime Android application development environments, collaborative software engineering workflows, professional debugging practices, and modern AI-assisted development methodologies.

One of the most important reflections from the internship is the realization that successful software engineers require a balanced combination of technical expertise, problem-solving abilities, communication skills, teamwork, adaptability, discipline, and continuous learning. While programming and technical implementation form the core foundation, qualities such as consistency, accountability, curiosity, professionalism, and collaborative mindset play an equally important role in long-term professional success.

- The internship highlighted the importance of practical implementation and realtime development experience beyond theoretical classroom learning.
 - Exposure to collaborative Android development workflows improved understanding of project coordination, version control systems, debugging
-

CHAPTER 5

CONCLUSION

The internship at MindMatrix was a highly valuable and professionally enriching experience that played a significant role in bridging the gap between academic learning and practical industry exposure. Throughout the internship period, continuous involvement in Android application development, Firebase integration, Generative AI assisted workflows, and realtime software engineering activities provided practical knowledge and technical experience beyond traditional classroom learning. The internship created an opportunity to understand modern software development practices, collaborative workflows, and professional expectations followed within the technology industry.

During the internship, practical exposure was gained in Android development technologies including Kotlin, Jetpack Compose, Firebase Authentication, Firebase Firestore, GitHub version control systems, and cloud-based backend services. The successful development of multiple Android applications such as Greeting Card App, Dice Roller App, Lemonade App, Business Card App, Art Space App, and Siri-Dhanya Hub App significantly improved understanding of responsive UI design, navigation systems, state management, realtime synchronization, debugging methodologies, and modular Android architecture. These projects strengthened implementation-oriented learning and improved confidence in handling realtime Android development tasks independently.

The internship also provided valuable exposure to Generative AI assisted development methodologies and modern software engineering practices. Working with AI-assisted tools helped improve productivity, debugging efficiency, frontend design workflows, and implementation support systems. Exposure to realtime project development activities improved understanding of structured coding practices, backend integration, Firebase cloud connectivity, application testing, and collaborative development workflows. The experience also strengthened logical thinking, analytical reasoning, debugging capabilities, and systematic problem-solving approaches while handling various implementation challenges and technical issues during project development.

Beyond technical learning, the internship contributed significantly towards professional growth and personal development. Continuous interaction with mentors and collaborative project discussions improved communication skills, teamwork abilities, documentation practices, and professional confidence. The internship highlighted the importance of discipline, accountability, adaptability, time management, and continuous learning within professional software engineering environments. It also demonstrated that successful software development requires both technical expertise and strong interpersonal skills for effective collaboration and project execution.

MindMatrix provided a supportive and innovation-driven learning environment that encouraged practical implementation, technical exploration, and industry-oriented skill development. The mentorship and guidance provided throughout the internship helped strengthen technical knowledge and career readiness for future software engineering opportunities. Exposure to modern Android development frameworks, Firebase technologies, and AI-assisted workflows also reinforced long-term career interest in Android application development, cloud technologies, Generative AI integration, and modern software engineering practices.

In conclusion, the internship at MindMatrix was not only an academic requirement but also an important professional learning experience that significantly improved technical competence, practical implementation skills, problem-solving abilities, and career preparedness. The knowledge gained, projects completed, and professional exposure achieved during the internship will continue to serve as a strong foundation for future academic growth and software engineering career opportunities.

CHAPTER 6

REFERENCES

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- Cloud Firestore Android SDK —
<https://firebase.google.com/docs/firestore/quickstart>
- Navigation Compose Guide —
<https://developer.android.com/guide/navigation/navigation-compose>
- Material Design 3 — <https://m3.material.io/>
- Kotlin Coroutines Documentation — <https://kotlinlang.org/docs/coroutines-overview.html>
- Firebase Free Tier Limits — <https://firebase.google.com/pricing>
- Kotlin Language Documentation — <https://kotlinlang.org/docs/home.html>